

Closed book portion:

- 1) Maxwell's Equations (differential)
- 2) Go from differential form to integral form (divergence theorem and Stokes' law)
- 3) Find E-field for simple geometry (plane) (Gauss' law)
- 4) Separation of variables in Cartesian coordinates and find the solutions to ODE
- 5) Find potential of a capacitor
- 6) Curl, divergence, and gradient
 - i) Can a function be an E-field? ($\text{curl} = 0 \Rightarrow$ it could be)

Open book portion:

- 1) Coulomb's Law
- 2) Potential
- 3) Boundary value and Fourier (write a_n and b_n)
- 4) Legendre polynomial (as 5th polynomial)

CO₂:



H₂O:

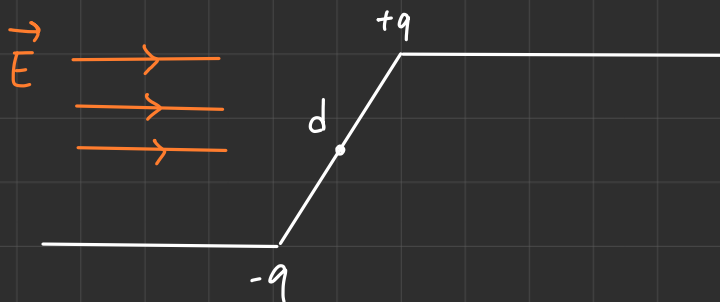


SF₆:

$$\vec{p} = \alpha_{\perp} \vec{E}_{\perp} + \alpha_{\parallel} \vec{E}_{\parallel}$$

$$\vec{p}_x = \alpha_{xx} \vec{E}_x + \alpha_{xy} \vec{E}_y + \alpha_{xz} \vec{E}_z$$

$$\vec{p}_y = \alpha_{yx} \vec{E}_x + \alpha_{yy} \vec{E}_y + \alpha_{yz} \vec{E}_z$$



Suppose $\vec{E}$ uniform

$$\sum \vec{F} = \sum \vec{E}q = \vec{E} \sum q = 0$$

$$\vec{N} = \left(\frac{\vec{d}}{2} \times q\vec{E} \right) + \left(-\frac{\vec{d}}{2} \times q\vec{E} \right) = q\vec{d} \times \vec{E} = \underbrace{\vec{p}}_{\text{dipole moment}} \times \vec{E}$$

What if E-field is non-uniform?

$$\sum \vec{F} = \vec{F}_+ + \vec{F}_- = q(\vec{E}_+ - \vec{E}_-) = q\Delta \vec{E}$$

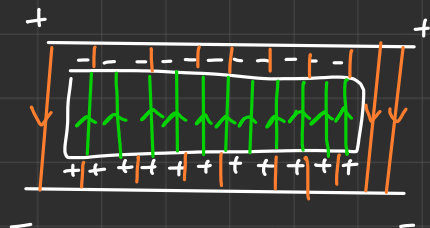
$$\Delta E_x = \nabla E_x \cdot \vec{d}$$

$$\Delta E = (\vec{d} \cdot \vec{\nabla}) \vec{E}$$

$$\Rightarrow \vec{F} = q\Delta \vec{E} = (q\vec{d} \cdot \vec{\nabla}) \vec{E} = (\vec{p} \cdot \vec{\nabla}) \vec{E}$$

Define

$$\vec{P} \equiv \frac{\text{dipole moment}}{\text{unit volume}}$$



Then $\vec{p} = \vec{P} d\tau$

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$$

Atoms
↓
Molecules
↓
Media

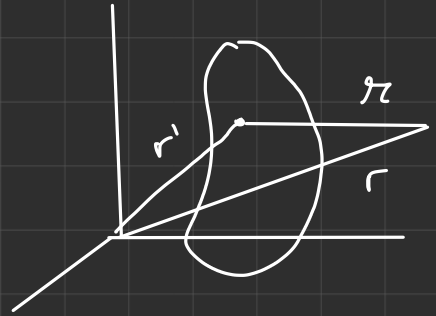
For multiple dipoles:

$$V(r) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\vec{P}(r') \cdot \hat{r}}{r^2} d\tau'$$

$$\nabla\left(\frac{1}{r}\right) = -\frac{\hat{r}}{r^2}$$

$$\vec{r} = \vec{r} - \vec{r}' \quad \text{and} \quad \nabla'\left(\frac{1}{r}\right) = \frac{\hat{r}}{r^2}$$

$$\Rightarrow V(r) = \frac{1}{4\pi\epsilon_0} \int_V \vec{P}(r') \cdot \nabla'\left(\frac{1}{r}\right) d\tau'$$



Identity: $\vec{\nabla} \cdot (f\vec{A}) = f(\vec{\nabla} \cdot \vec{A}) + \vec{A} \cdot (\vec{\nabla} f)$

Thus, $\vec{\nabla}' \cdot \left(\frac{1}{r} \vec{P}(r')\right) = \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}(r')) + \vec{P}(r') \cdot (\vec{\nabla}'\left(\frac{1}{r}\right))$

$$\begin{aligned} \Rightarrow V(r) &= \frac{1}{4\pi\epsilon_0} \left[\int_V \vec{\nabla}' \cdot \left(\frac{\vec{P}(r')}{r}\right) d\tau' - \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau' \right] \\ &= \frac{1}{4\pi\epsilon_0} \oint_S \frac{1}{r} \vec{P}(r') \cdot d\vec{a} - \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{r} (\vec{\nabla}' \cdot \vec{P}) d\tau' \end{aligned}$$

Define:

$$\sigma_b \equiv \vec{P} \cdot \hat{n} \quad (\text{surface charge})$$

$$\rho_b \equiv -\vec{\nabla}' \cdot \vec{P} \quad (\text{volume charge})$$

$$V(r) = \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma_b}{r} da + \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho_b}{r} d\tau'$$

Midterm 1 Closed Book Solution:

1) We want $E = 0$ at $(3, 1, 1)$

$$\begin{aligned}\vec{r}_1 &= 3\hat{x} + \hat{y} + \hat{z} & |r_1| &= \sqrt{11} \\ \vec{r}_2 &= (3-2)\hat{x} + \hat{y} + \hat{z} = \hat{x} + \hat{y} + \hat{z} & |r_2| &= \sqrt{3}\end{aligned}$$

$$\hat{r}_1 = \frac{3\hat{x} + \hat{y} + \hat{z}}{\sqrt{11}} \quad ; \quad -\hat{r}_2 = \frac{\hat{x} + \hat{y} + \hat{z}}{\sqrt{3}}$$

$$\text{Then } \vec{E} = \frac{1}{4\pi\epsilon_0} \left[\frac{3\hat{x} + \hat{y} + \hat{z}}{11^{3/2}} + \frac{\hat{x} + \hat{y} + \hat{z}}{3^{3/2}} \right]$$

But we only want the x component of E

$$\Rightarrow \vec{E}_x = \frac{1}{4\pi\epsilon_0} \left(q_1 \frac{3x}{11^{3/2}} + q_2 \frac{x}{3^{3/2}} \right) = 0$$

$$\Rightarrow q_2 = - \frac{3q_1 (3^{3/2})}{11^{3/2}}$$

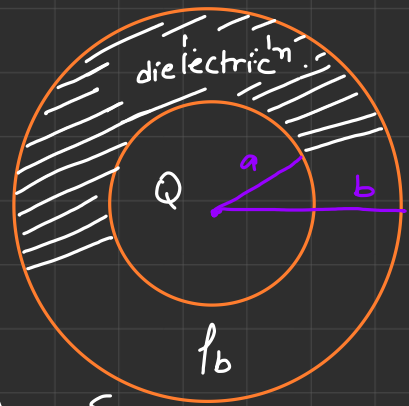
$$= - \frac{3^{5/2}}{11^{3/2}} q_1$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$= \epsilon_0 \vec{E} + \epsilon_0 \chi \vec{E}$$

$$= \epsilon_0 (1 + \chi) \vec{E}$$

$$= \epsilon \vec{E} \quad ; \text{ where } \epsilon = \epsilon_0 (1 + \chi)$$



Define the relative permeability $\epsilon_r = (1 + \chi) = \frac{\epsilon}{\epsilon_0}$

• $r < a$: $\vec{E} = \vec{P} = \vec{D} = \vec{0}$

• $r > a$: $\int \vec{D} \cdot d\vec{a} = Q_f$
 $4\pi r^2 D = Q \Rightarrow \vec{D} = \frac{Q}{4\pi r^2} \hat{r}$

$$\vec{D} = \epsilon \vec{E}$$

$$\Rightarrow \vec{E} = \begin{cases} \frac{Q}{4\pi \epsilon r^2} \hat{r} & ; a < r < b \\ \frac{Q}{4\pi \epsilon_0 r^2} \hat{r} & ; r > b \end{cases}$$

Compute V at $r = 0$

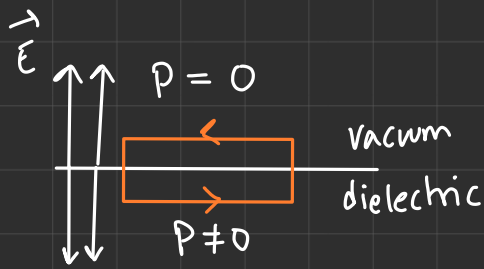
$$V(0) = - \int_{\infty}^0 \vec{E} \cdot d\vec{l}$$

$$= - \int_{\infty}^b \frac{Q}{4\pi \epsilon_0 r^2} dr - \int_b^a \frac{Q}{4\pi \epsilon r^2} dr - \int_a^0 0 dr$$

$$= \left[\frac{Q}{4\pi \epsilon_0 r} \right]_{\infty}^b + \left[\frac{Q}{4\pi \epsilon r} \right]_b^a$$

$$= \frac{Q}{4\pi} \left[\frac{1}{\epsilon_0 b} + \frac{1}{\epsilon a} - \frac{1}{\epsilon b} \right]$$

We can also find the potential from Laplace's Equation:



$$\nabla \times \mathbf{P} \stackrel{?}{=} 0 \quad \text{No}$$

$$\oint \mathbf{P} \cdot d\mathbf{l} \neq 0$$

$$\oint \mathbf{P} \cdot d\mathbf{l} \stackrel{?}{=} 0$$

$$P_x = \epsilon_0 (\chi_{xx} E_x + \chi_{xy} E_y + \chi_{xz} E_z)$$

$$P_y = \epsilon_0 (\chi_{yx} E_x + \chi_{yy} E_y + \chi_{yz} E_z)$$

$$P_z = \epsilon_0 (\chi_{zx} E_x + \chi_{zy} E_y + \chi_{zz} E_z)$$

$$E_a^\perp - E_b^\perp = \sigma$$

$$E_a'' - E_b'' = 0$$

$$D_a^\perp - D_b^\perp = \sigma_f$$

$$\epsilon_a E_a^\perp - \epsilon_b E_b^\perp = \sigma_f$$

$$\Rightarrow \epsilon_a \frac{\partial V}{\partial r} - \epsilon_b \frac{\partial V_b}{\partial r} = -\sigma_f$$

$$V_a = V_b$$



$$E^\perp$$

$$\nabla \cdot E = \frac{\rho}{\epsilon_0} \quad \oint E \cdot da = q_{enc}$$

$$E''$$

$$\nabla \times E = 0 \quad \oint E \cdot dl$$

$$V_a, V_b$$

$$V_a = V_b$$

$$\sigma$$

No dielectric: $E_a^\perp - E_b^\perp = \sigma$

With dielectric: $D_a^\perp - D_b^\perp = \sigma_f \Leftrightarrow \epsilon_a E_a^\perp - \epsilon_b E_b^\perp = \sigma_f$

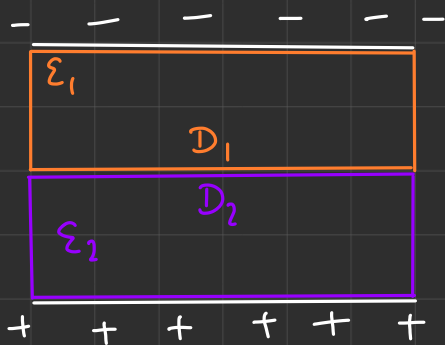
$$\Leftrightarrow \epsilon_a \frac{\partial V_a}{\partial n} - \epsilon_b \frac{\partial V_b}{\partial n} = -\sigma_f$$

$$V = - \int \vec{E} \cdot d\vec{l}$$

True even with dielectrics

~~$$V' = - \int \vec{D} \cdot d\vec{l}$$~~

Not true in general

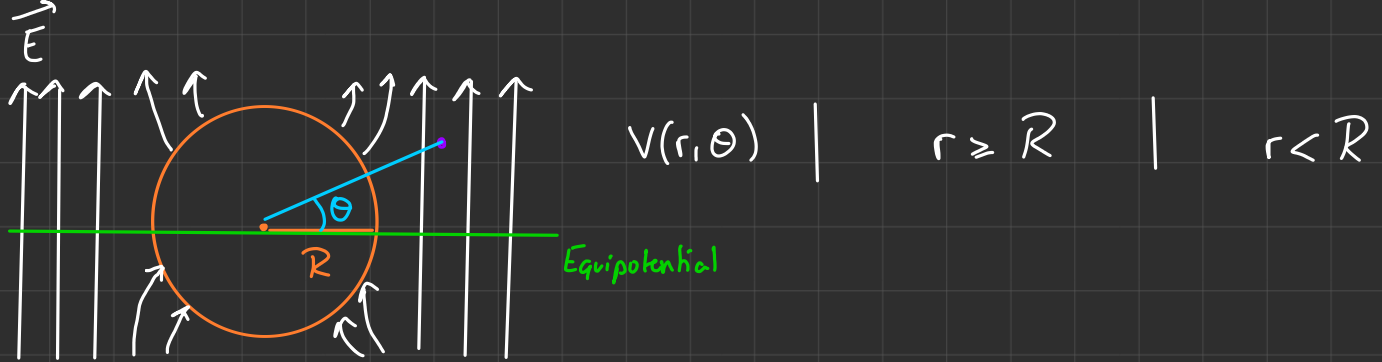


Boundary Conditions:

① $\epsilon_a E_a^\perp - \epsilon_b E_b^\perp = \sigma_f$

② $V_a = V_b$

③ $D_a'' - D_b'' = P_a'' - P_b''$



i) $V_{in} = V_{out}$

ii) $\epsilon \frac{\partial V_{in}}{\partial n} = \epsilon_0 \frac{\partial V_{out}}{\partial n}$

iii) $V = -E_0 r \cos(\theta)$ for $r \gg R$

Spherical geometry with symmetries has solution of Legendre poly.

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos(\theta))$$

• For $r < R$: $V(r, \theta) = \sum_{l=0}^{\infty} A_l r^l P_l(\cos(\theta))$

• For $r > R$: $V(r, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{r^{l+1}} P_l(\cos(\theta)) - E_0 r \cos(\theta)$

Apply boundary conditions:

i) $V_{in} = V_{out} \Leftrightarrow \sum_{l=0}^{\infty} A_l R^l P_l(\cos(\theta)) = \sum_{l=0}^{\infty} \frac{B_l}{R^{l+1}} P_l(\cos(\theta)) - E_0 R \cos(\theta)$

For $l=1$: $A_1 R = \frac{B_1}{R^2} - E_0 R$

For $l \neq 1$: $A_l R^l = \frac{B_l}{R^{l+1}}$

Take derivative: $\epsilon_R \sum_{l=0}^{\infty} l A_l R^{l-1} P_l(\cos(\theta)) = - \sum_{l=0}^{\infty} \frac{(l+1) B_l}{R^{l+2}} P_l(\cos(\theta)) - E_0 \cos(\theta)$

For $l=1$: $\epsilon_r A_1 = -\frac{2B_1}{R^3} - E_0$

For $l \neq 1$: $\epsilon_r l A_l R^{l-1} = -\frac{(l+1)B_l}{R^{l+2}}$

If $l \neq 1$: $A_l = B_l = 0$

$\epsilon_r = \frac{\epsilon}{\epsilon_0}$

If $l=1$: $A_1 = -\frac{3}{\epsilon_r + 2} E_0$ $B_1 = \frac{\epsilon_r - 1}{\epsilon_r + 2} R^3 E_0$

$$V_{in} = -\frac{3E_0}{\epsilon_r + 2} r \cos(\theta) = -\frac{3E_0}{\epsilon_r + 2} \hat{z}$$

$$\vec{E} = -\nabla V = \frac{3}{\epsilon_r + 2} E_0$$

Calculating the energy stored in a capacitor



$$W = \frac{1}{2} C V^2$$

$$W = \frac{\epsilon_0}{2} \int_V E^2 d\tau =$$

If a dielectric is present: $W = \frac{\epsilon_0}{2} \int \epsilon_r E^2 d\tau$

$$= \frac{1}{2} \int \vec{D} \cdot \vec{E} d\tau$$

$\Delta W = \int \Delta \rho_f V d\tau$ Green's Reciprocity Theorem

$\uparrow$ free charge density $\uparrow$ potential

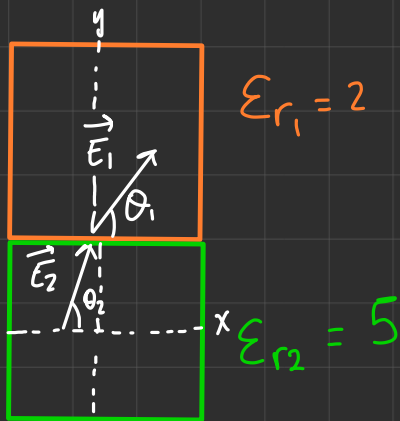
$$\vec{\nabla} \cdot \vec{D} = \rho_f \Rightarrow \Delta (\vec{\nabla} \cdot \vec{D}) = \vec{\nabla} \cdot (\Delta \vec{D}) = \Delta \rho_f$$

$$\Rightarrow \Delta W = \int \underbrace{\vec{\nabla} \cdot (\Delta \vec{D})}_V V d\tau = \int \underbrace{\vec{\nabla} \cdot [(\Delta \vec{D}) V]}_{\text{Apply div theorem and it goes to zero over all space}} d\tau + \int \Delta \vec{D} \cdot \vec{E} d\tau$$

$$\vec{\nabla} \cdot [(\Delta \vec{D}) V] = [\vec{\nabla} \cdot (\Delta \vec{D})] V + \Delta \vec{D} \cdot \underbrace{\vec{\nabla} V}_{-\vec{E}}$$

$$\frac{1}{2} \Delta(\vec{D} \cdot \vec{E}) = \frac{1}{2} \Delta(\epsilon E^2) \xrightarrow{\text{derivative}} \epsilon \Delta \vec{E} \cdot \vec{E} = \Delta \vec{D} \cdot \vec{E}$$

$$\Rightarrow W = \frac{1}{2} \int \vec{D} \cdot \vec{E} d\tau$$



$$\vec{E}_1 = 2\hat{x} - 3\hat{y} + 5\hat{z} \quad (\text{V/m})$$

$$\vec{D}_1 = \epsilon_0 \epsilon_r \vec{E}_1 = 4\epsilon_0 \hat{x} - 6\epsilon_0 \hat{y} + 10\epsilon_0 \hat{z}$$

Boundary Conditions:

$$\perp : \vec{\nabla} \cdot \vec{D} = \rho_f$$

$$\parallel : \vec{\nabla}_x \vec{D} = \vec{\nabla}_x \vec{P}$$

$$\begin{cases} E_{2x} = E_{1x} = 2 \\ E_{2y} = E_{1y} = -3 \\ E_{2z} = ? \end{cases}$$

$$D_{2x} = ?$$

$$D_{2y} = ?$$

$$D_{2z} = D_{1z} = 10\epsilon_0$$

$$\Rightarrow \begin{cases} \vec{E}_2 = 2\hat{x} - 3\hat{y} + E_{2z}\hat{z} \\ \vec{D}_2 = \underline{D_{2x}}\hat{x} + \underline{D_{2y}}\hat{y} + \underline{10\epsilon_0}\hat{z} \end{cases}$$

$$\vec{D}_2 = \epsilon_0 \epsilon_{r2} \vec{E}_2 = \underline{10\epsilon_0}\hat{x} - \underline{15\epsilon_0}\hat{y} + \underline{5\epsilon_0 E_{2z}}\hat{z}$$

$$\Rightarrow \begin{cases} D_{2x} = 10\epsilon_0 \\ D_{2y} = -15\epsilon_0 \\ 5\epsilon_0 E_{2z} = 10\epsilon_0 \Rightarrow E_{2z} = 2 \end{cases}$$

$$\Rightarrow \begin{cases} \vec{E}_2 = 2\hat{x} - 3\hat{y} + 2\hat{z} \\ \vec{D}_2 = 10\epsilon_0 \hat{x} - 15\epsilon_0 \hat{y} + 10\epsilon_0 \hat{z} \end{cases}$$

$$\begin{cases} \vec{E}_1 \cdot \hat{z} = |E_1| \sin(\theta_1) = \sqrt{38} \sin(\theta_1) = 5 \Rightarrow \theta_1 = 54.2^\circ \\ \vec{E}_2 \cdot \hat{z} = |E_2| \sin(\theta_2) = \sqrt{17} \sin(\theta_2) = 2 \Rightarrow \theta_2 = 29^\circ \end{cases}$$

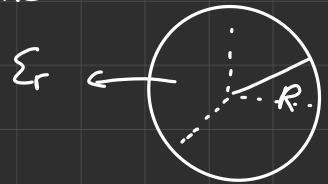
$$\cdot \tan(\theta_1) = \frac{E_{1z}}{\sqrt{E_{1x}^2 + E_{1y}^2}} = \frac{D_{1z}/\epsilon_0 \epsilon_{r1}}{\sqrt{E_{1x}^2 + E_{1y}^2}}$$

$$\cdot \tan(\theta_2) = \frac{E_{2z}}{\sqrt{E_{2x}^2 + E_{2y}^2}} = \frac{D_{2z}/\epsilon_0 \epsilon_{r2}}{\sqrt{E_{2x}^2 + E_{2y}^2}}$$

$$\Rightarrow \frac{\tan(\theta_1)}{\tan(\theta_2)} = \frac{\epsilon_{r2}}{\epsilon_{r1}}$$

Example 4.9: The energy in a uniform sphere

$$q_{\text{enc}} = \rho_f \frac{4}{3} \pi R^3$$



Gauss' Law:

$$\cdot \text{For } r < R: \int \vec{D} \cdot d\vec{a} = q_{\text{enc}}$$

$$\vec{D} (4\pi r^2) = \rho_f \frac{4}{3} \pi r^3$$

$$\Rightarrow \vec{D} = \frac{\rho_f}{3} r \hat{r}$$

$$\cdot \text{For } r > R: \int \vec{D} \cdot d\vec{a} = q_{\text{enc}}$$

$$\vec{D} (4\pi r^2) = \rho_f \frac{4}{3} \pi R^3$$

$$\Rightarrow \vec{D} = \frac{\rho_f}{3} \frac{R^3}{r^2} \hat{r}$$

$$\Rightarrow \vec{E}(r) = \begin{cases} \frac{\rho_f}{3\epsilon_0 \epsilon_r} r \hat{r} & ; r < R \\ \frac{\rho_f}{3\epsilon_0} \frac{R^3}{r^2} \hat{r} & ; r > R \end{cases}$$

Method 1: $\int E^2$

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau$$

$$= \frac{\epsilon_0}{2} \left[\left(\frac{\rho_f}{3\epsilon_0 \epsilon_r} \right)^2 \int_0^R \int_0^\pi \int_0^{2\pi} r^2 dr d\phi d\theta + \left(\frac{\rho_f}{3\epsilon_0} \right)^2 R^6 \int_R^\infty \int_0^\pi \int_0^{2\pi} \frac{r^2}{r^4} dr d\phi d\theta \right]$$

$$= \frac{2\pi}{9\epsilon_0} \rho_f^2 R^5 \left(\frac{1}{5\epsilon_r^2} + 1 \right)$$

Method 2: $\int \vec{D} \cdot \vec{E}$

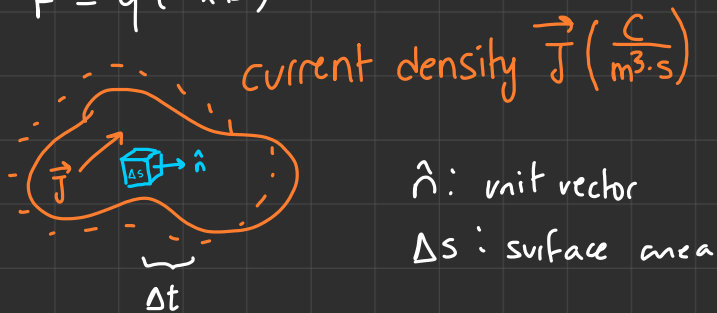
$$W = \int \vec{D} \cdot \vec{E} \, d\tau = \frac{2\pi}{9\epsilon_0} \rho_p^2 R^5 \left(\frac{1}{5\epsilon_r} + 1 \right)$$

Magnetostatics

It is the study of E&M when we have **Steady currents**.

Current (I): flow of electrons ($\frac{\text{Coulomb}}{\text{sec}}$)

$$\vec{F} = q(\vec{v} \times \vec{B})$$



$$\vec{J} \cdot \hat{n} \Delta s = \frac{\Delta q}{\Delta t}$$

$$\Delta q = \rho \vec{v} \cdot \hat{n} \Delta s \Delta t$$

$$V \Delta t = \Delta l$$

$$V \Delta t \Delta s = \Delta V \leftarrow \text{volume}$$

$$\Rightarrow \underline{\vec{J} \cdot \hat{n} \Delta s \Delta t} = \underline{\rho \vec{v} \cdot \hat{n} \Delta s \Delta t}$$

$$\Rightarrow \boxed{\vec{J} = \rho \vec{v}} = Nq \vec{v}$$

$$I = \int \vec{J} \cdot \hat{n} ds = - \frac{dQ}{dt} = - \frac{d}{dt} \int \rho d\tau$$

Also $\int_S \vec{J} \cdot \hat{n} ds = \int_S \vec{J} \cdot d\vec{s} = \int_V \nabla \cdot \vec{J} d\tau$

$$\Rightarrow \boxed{\nabla \cdot \vec{J} = - \frac{\partial \rho}{\partial t}} \quad \text{Advection Equation (continuity)}$$

Suppose a horseshoe magnet

$$F = q (\vec{v} \times \vec{B})$$

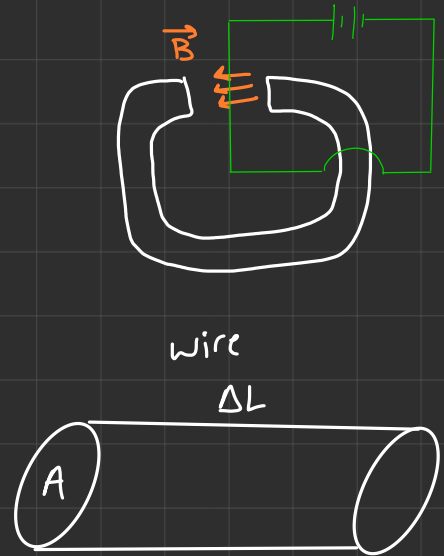
$$\Delta F = N \Delta V q \vec{v} \times \vec{B}$$

$$\text{and } \vec{J} = Nq \vec{v}$$

$$\Rightarrow \Delta F = \vec{J} \times \vec{B} \Delta V$$
$$= \vec{I} \times \vec{B} \Delta L$$

$$\Rightarrow \frac{\Delta F}{\Delta L} = \vec{I} \times \vec{B}$$

$$\Rightarrow F \propto \text{current}$$



Maxwell's Equations:

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$c^2 \vec{\nabla} \times \vec{B} = \frac{\partial \vec{E}}{\partial t} + \frac{\vec{J}}{\epsilon_0}$$

→ Take divergence

$$c^2 \vec{\nabla} \cdot (\vec{\nabla} \times \vec{B}) = \frac{1}{\epsilon_0} \vec{\nabla} \cdot \vec{J}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} = 0$$

This means current loops generate magnetic fields in magnetostatics

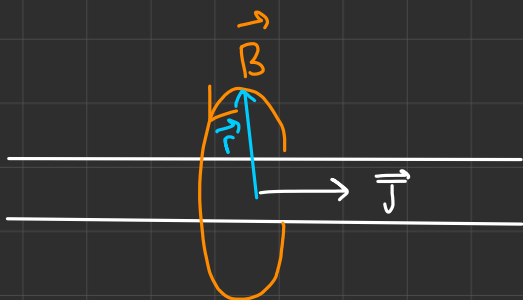
$$\oint_S (\vec{\nabla} \times \vec{B}) \cdot \hat{n} ds = \int_C \vec{B} \cdot d\vec{s}$$

$$\parallel$$

$$\frac{1}{c^2 \epsilon_0} \int \vec{J} \cdot \hat{n} ds = \frac{1}{c^2 \epsilon_0} I$$

$$\Rightarrow \boxed{\int_C \vec{B} \cdot d\vec{s} = \frac{I}{c^2 \epsilon_0}} \quad \text{Ampère's Law}$$

Magnetic Field of an Infinite Wire



$$\oint \vec{B} \cdot d\vec{s} = B(2\pi r)$$

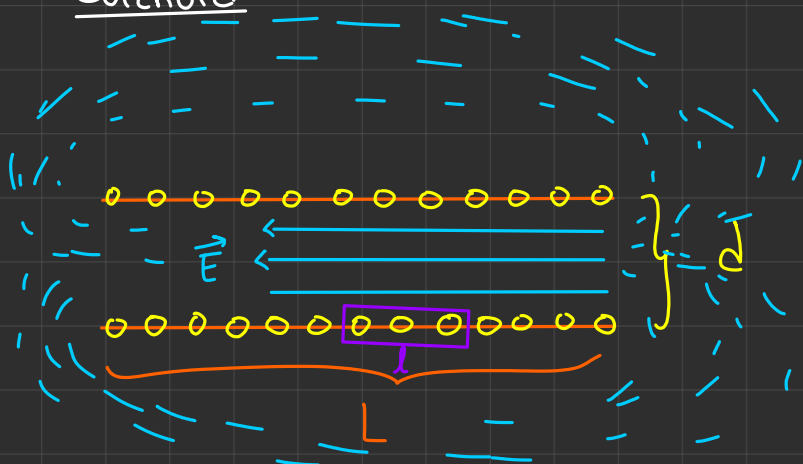
$$= \frac{I}{\epsilon_0 c^2}$$

$$\Rightarrow B = \frac{1}{2\pi \epsilon_0 c^2} \frac{I}{r}$$

$$\vec{B} = \frac{1}{4\pi \epsilon_0 c^2} \frac{2 \vec{I} \times \hat{e}_r}{r}$$

$$\frac{1}{4\pi \epsilon_0 c^2} = 10^{-7} \text{ mks}$$

Solenoid:



$$L \gg d$$

$$B_{in} \gg B_{out}$$

Given I and N

$$\int \vec{B} \cdot d\vec{s} = Bl = N \frac{I}{\epsilon_0 c^2}$$

$$\Rightarrow B = \frac{IN}{\epsilon_0 c^2 L} = \frac{nI}{\epsilon_0 c^2} ; n = \frac{N}{L}$$



$$\vec{F} = q(\vec{v} \times \vec{B}) + q\vec{E}$$

S: particle moves w.r.t wire

S': wire moves w.r.t particle

$$\vec{F} = q\vec{v} \times \vec{B} = \frac{q}{4\pi\epsilon_0} =$$

$$\vec{B} =$$

$$\vec{I} =$$

Biot-Savart Law

stationary charges $\rightarrow$ electric fields

$$\frac{\partial \rho}{\partial t} = 0$$

stationary current $\rightarrow$ magnetic fields

$$\frac{\partial \vec{J}}{\partial t} = 0$$

$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \Rightarrow \nabla \cdot \vec{J} = 0$$

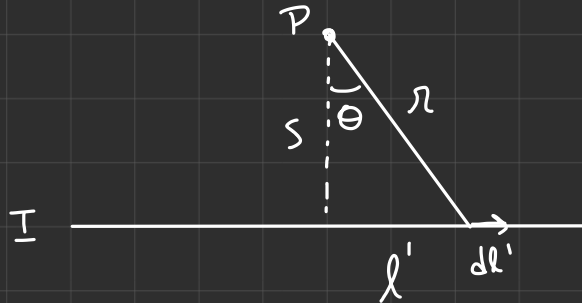
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r') \hat{r}}{r^2} dr' \quad ; \quad \epsilon_0 = 8.85 \times 10^{-12} \text{ C/V.m}^2$$

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\vec{I} \times \hat{r}}{r^2} dl' = \frac{\mu_0 I}{4\pi} \int \frac{dl' \times \hat{r}}{r^2} \quad ; \quad \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

$$1 \text{ Tesla} = 10^4 \text{ Gauss}$$

$$\downarrow$$

$$\left[\frac{\text{N}}{\text{A.m}} \right]$$



$$dl' \times \vec{r} = dl' r \sin(90^\circ - \theta) = dl' r \cos(\theta)$$

$$l' = s \tan(\theta)$$

$$dl' = \frac{s}{\cos^2(\theta)} d\theta$$

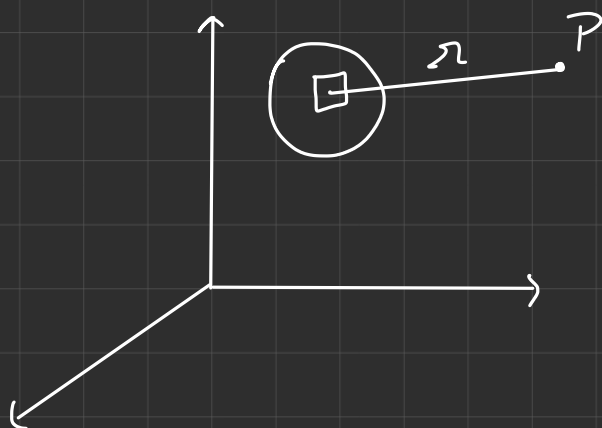
$$\frac{1}{r^2} = \frac{\cos^2(\theta)}{s^2}$$

$$\Rightarrow \vec{B} = \frac{\mu_0 I}{4\pi} \int_{\theta_1}^{\theta_2} \frac{\cos^2(\theta)}{s^2} \frac{s}{\cos^2(\theta)} \cos(\theta) d\theta = \frac{\mu_0 I}{4\pi s} \int_{\theta_1}^{\theta_2} \cos(\theta) d\theta$$

$$= \frac{\mu_0 I}{4\pi s} (\sin(\theta_2) - \sin(\theta_1))$$

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{k}(\vec{r}') \times \hat{r}}{r^2} d\vec{r}'$$

$$= \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}') \times \hat{r}}{r^2} d\tau'$$



$$\vec{B}(x, y, z)$$

$$\vec{J}(x', y', z')$$

$$\vec{r} = (x - x') \hat{x} + (y - y') \hat{y} + (z - z') \hat{z}$$

$$d\tau' = dx' dy' dz'$$

$$\nabla \cdot \vec{B} = \frac{\mu_0}{4\pi} \int \nabla \cdot (\vec{J} \times \frac{\hat{r}}{r^2}) d\tau'$$

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

$$\nabla \cdot (\vec{J} \times \frac{\hat{r}}{r^2}) = \frac{\hat{r}}{r^2} \cdot (\underbrace{\nabla \times \vec{J}}_{=0}) - \vec{J} \cdot (\underbrace{\nabla \times \frac{\hat{r}}{r^2}}_{=0})$$

ABSENT

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} \Rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \nabla(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$$\text{If } \nabla \cdot \vec{A} = 0 \Rightarrow \nabla^2 \vec{A} = -\mu_0 \vec{J}$$

$$\Rightarrow \vec{A} = \vec{A}_0 + \nabla \lambda$$

$$\text{Then } \nabla \cdot \vec{A} = \nabla \cdot \vec{A}_0 + \nabla^2 \lambda = 0$$

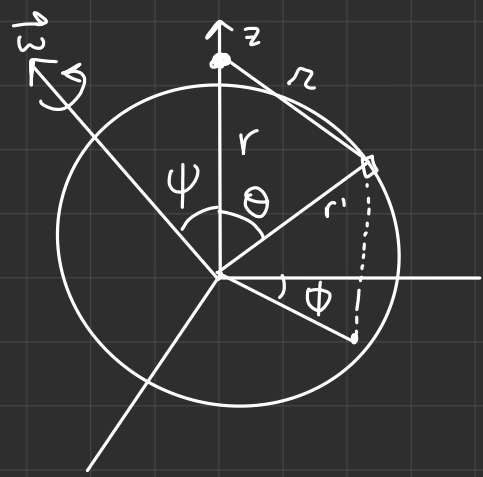
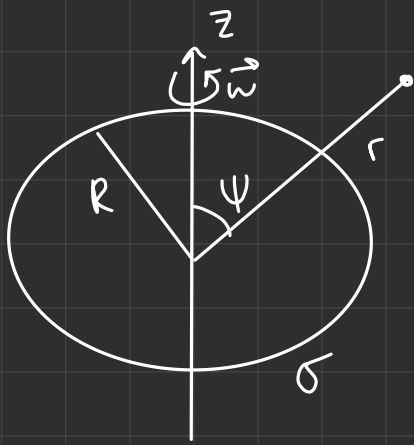
$$\Rightarrow \nabla^2 \lambda = -\nabla \cdot \vec{A}_0 \quad (\text{Poisson's Equation})$$

$$\exists \lambda \text{ s.t. } \nabla \cdot \vec{A}_0 = 0 \text{ always}$$

- If $f \rightarrow 0$ as $r \rightarrow \infty$, solution can be written
- If $\nabla \cdot \vec{A}_0 \rightarrow 0$ as $r \rightarrow \infty$, $\lambda = \frac{1}{4\pi} \int \frac{\nabla \cdot \vec{A}_0}{r} d\tau$

$$\text{From } \nabla^2 \vec{A} = -\mu_0 \vec{J}$$

$$\text{If } \vec{J} \rightarrow 0 \text{ as } r \rightarrow \infty, \text{ then } \vec{A}(r) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(r')}{r} d\tau'$$



$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{\vec{K}}{r} da$$

$$\vec{K} = \sigma \vec{v}$$

$$\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \omega \sin(\psi) & 0 & \omega \cos(\psi) \\ R \sin(\theta) \cos(\phi) & R \sin(\theta) \sin(\phi) & R \sin(\theta) \end{vmatrix}$$

$$= -\hat{x} R \omega \sin(\theta) \cos(\phi) \cos(\psi) - \hat{y} [\omega R \sin(\theta) \sin(\psi) - \omega R \sin(\theta) \cos(\phi) \cos(\psi)]$$

$$+ \hat{z} [\omega R \sin(\theta) \sin(\phi) \sin(\psi)]$$

$$= -\hat{y} (\sin(\psi) \cos(\theta)) R \omega$$

$$r = \sqrt{R^2 + r^2 - 2Rr \cos(\theta)}$$

$$da = R^2 \sin(\theta) d\theta d\phi$$

$$\text{Then } \vec{A} = \frac{-\mu_0}{4\pi} \int \frac{\sigma \sin(\psi) \cos(\theta) R^2 R \omega \sin(\theta)}{\sqrt{R^2 + r^2 - 2Rr \cos(\theta)}} d\phi d\theta$$

$$= -\frac{\mu_0 R^3 \sigma \sin(\psi) \omega (2\pi)}{4\pi} \int_0^\pi \frac{\cos(\theta) \sin(\theta)}{\sqrt{R^2 + r^2 - 2Rr \cos(\theta)}} d\theta$$

$$= -\frac{\mu_0 R^3 \sigma \sin(\psi) \omega}{2} \left(-\int_1^0 \frac{u du}{\sqrt{R^2 + r^2 - 2Rru}} \right)$$

$$= -\frac{\mu_0 R^3 \sigma \sin(\psi) \omega}{2} \left[-\frac{R^2 + r^2 + Rru}{3R^2 r^2} \sqrt{R^2 + r^2 - 2Rru} \right]_1^0$$

$$= \frac{\mu_0 R^3 \sigma \sin(\psi) \omega}{2} \frac{1}{3R^2 r^2} \left[(R^2 + r^2 + Rr) |R-r| - (R^2 + r^2 - Rr)(R+r) \right]$$

$$= \frac{\mu_0 R^3 \sigma \omega \sin(\psi)}{2} \begin{cases} \frac{2r}{3R^2} & , \quad r < R \\ \frac{2R}{3r^2} & , \quad r > R \end{cases}$$

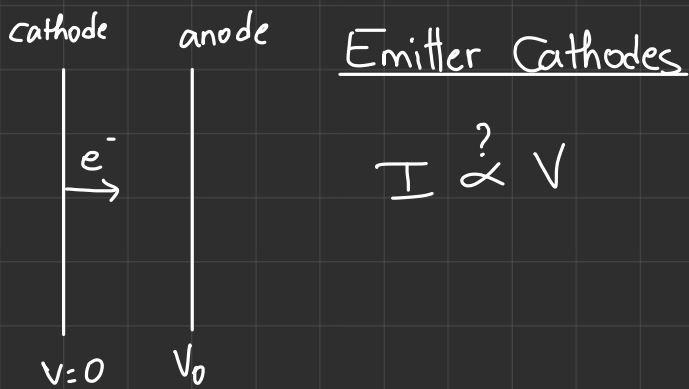
$$A(r) = \begin{cases} \frac{\mu_0 R \sigma}{3} (\vec{\omega} \times \vec{r}) & ; \quad r < R \\ \frac{\mu_0 R^4 \sigma}{3 r^3} (\vec{\omega} \times \vec{r}) & ; \quad r > R \end{cases}$$

The vector potential and current density mimic each other

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$\vec{B} = \frac{2\mu_0 R \omega \sigma}{3} \hat{z}$$

ABSENT



$$I = \left[\frac{C}{s} \right]$$

$$A = [m^2]$$

$$v = \left[\frac{m}{s} \right]$$

$$\frac{\partial^2 V}{\partial x^2} = -\frac{\rho}{\epsilon_0}$$

$$\rho = \frac{I}{V \cdot A}$$

$$\frac{dq}{dt} = A \rho \frac{dx}{dt}$$

$$I = A \rho v$$

$$d = \frac{1}{2} \frac{v^2}{a}$$

$$F = ma = qE$$

$$E = -\frac{V_0}{d}$$

$$\frac{d^2 V}{dx^2} = \beta v^{1/2} \quad ; \quad \beta = \frac{I}{\epsilon_0 A} \sqrt{\frac{m}{2q}}$$

$$\text{let } v' = \frac{dv}{dx}, \text{ then } \frac{dv'}{dx} = \beta v^{-1/2}$$

$$E = -\frac{\partial V}{\partial x}$$

$$v' \frac{dv'}{dx} = \beta v^{-1/2} v'$$

$$v' dv' = \beta v^{-1/2} dv$$

$$\text{Integrate: } \frac{(v')^2}{2} = 2\beta v^{1/2} + C$$

$$\bullet \text{ At } x=0, V=0: E=0 \text{ (conductor)}$$

$$\Rightarrow 0 = 0 + C \Rightarrow C = 0$$

$$\Rightarrow (V')^2 = 4\beta V^{\frac{1}{2}}$$

$$V' = 2\sqrt{\beta} V^{\frac{1}{4}}$$

$$\frac{dV}{dx} = 2\sqrt{\beta} V^{\frac{1}{4}}$$

$$V^{-\frac{1}{4}} dV = 2\sqrt{\beta} dx$$

$$\text{Integrate: } \frac{4}{3} V^{\frac{3}{4}} = 2\sqrt{\beta} x + C$$

$$\cdot \text{At } x=0, V=0: 0 = 0 + C \Rightarrow C=0$$

$$\text{Thus } V^{\frac{3}{4}} = \frac{3}{2} \sqrt{\beta} x$$

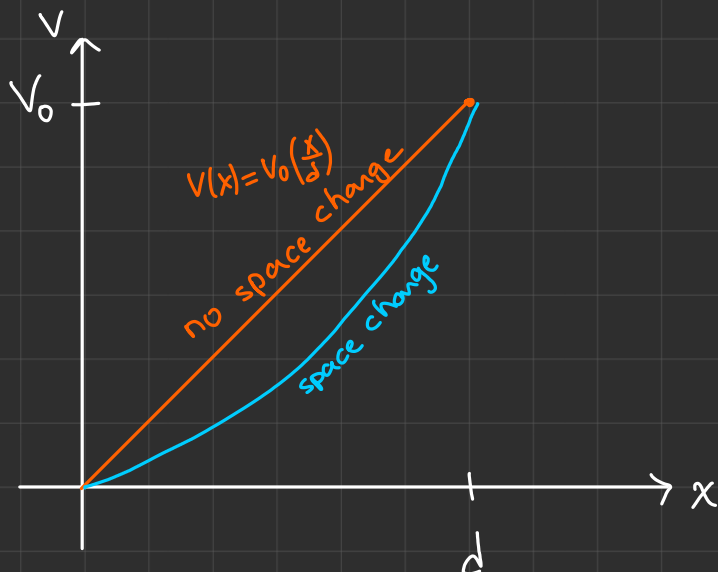
$$V = \left(\frac{3}{2} \sqrt{\beta} x \right)^{\frac{4}{3}} = \left(\frac{9}{4} \beta \right)^{\frac{2}{3}} x^{\frac{4}{3}}$$

$$V(x) = \left(\frac{81}{16} \frac{I^2}{\epsilon_0^2 A^2} \frac{m}{2q} \right)^{\frac{1}{3}} x^{\frac{4}{3}}$$

$$= \left(\frac{81m}{32\epsilon_0^2 A^2} \frac{I^2}{9} \right)^{\frac{1}{3}} x^{\frac{4}{3}}$$

$$V(d) = V_0 = [\dots]^{\frac{1}{3}} d^{\frac{4}{3}} \Rightarrow [\dots]^{\frac{1}{3}} = \frac{V_0}{d^{\frac{4}{3}}}$$

$$\text{Replace: } V(x) = V_0 \left(\frac{x}{d} \right)^{\frac{4}{3}}$$



- space charge is damping the potential
- $E = \frac{dV}{dx}$ is strongest at $x=d$

$$V_0 = \left(\frac{81m}{32\varepsilon_0^2 A^2} \frac{1}{q} \right)^{\frac{1}{3}} d^{\frac{4}{3}} I^{\frac{2}{3}}$$

$$V_0^3 = \left(\frac{81m}{32\varepsilon_0^2 A^2} \frac{1}{q} \right) d^4 I^2$$

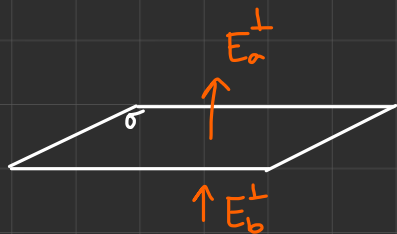
$$I^2 = \left(\frac{32\varepsilon_0^2 A^2 q}{81m} \right) \frac{V_0^3}{d^4}$$

$$I = \frac{4\varepsilon_0 A}{d^2} \left(\frac{2\sqrt{q}}{81m} \right)^{\frac{1}{2}} V_0^{\frac{3}{2}}$$

$$I \propto V^{\frac{3}{2}} \quad \text{Child's Law}$$

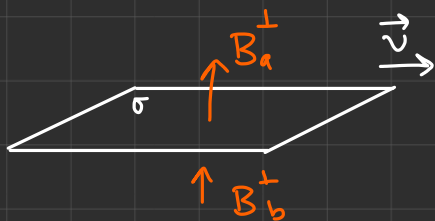
Boundary Conditions:

• Perpendicular: $\perp$



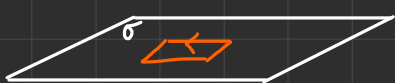
$$\int \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\varepsilon_0} \Rightarrow E_a^{\perp} - E_b^{\perp} = \frac{\sigma}{\varepsilon_0}$$

Now imagine this sheet of charge is moving



$$\int \vec{B} \cdot d\vec{a} = 0 \Rightarrow B_a^{\perp} = B_b^{\perp}$$

• Parallel: $\parallel$



$$\oint \vec{E} \cdot d\vec{l} = 0, \quad E_a^{\parallel} = E_b^{\parallel}$$

$$\Rightarrow E_a^{\parallel} - E_b^{\parallel} = \frac{\sigma}{\varepsilon} \hat{n}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I = (\mathcal{B}_a'' - \mathcal{B}_b'') l$$

$$= \mu_0 k l$$

$$\Rightarrow \mathcal{B}_a'' - \mathcal{B}_b'' = \mu_0 k$$

- $V_a = V_b$
- $\nabla \cdot \vec{A} = 0 \Rightarrow A_a = A_b$

$$A_{\text{dipole}} = \frac{\mu_0 I}{4\pi r^2} \oint r' \cos(\alpha) dl' , \quad \oint (\hat{r} \cdot r') dl' = -\hat{r} \times \underbrace{\int d\vec{a}'}_{\text{vector area}}$$

$$= \frac{\mu_0}{4\pi r^2} \frac{\vec{m} \times \vec{r}}{r^2} \quad \text{where } \vec{m} = I \int d\vec{a}$$

↓
Allows the study of magnetization

